Measuring Agenda-Setting Influence from Legislative Speech

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Motivation

Ken Clarke, MP for Rushcliffe

- Article 50 debate
- Focus on individual responsibility and conscience
- "A masterclass display of oratory and expertise."

Methodological argument

Agenda-setting influence: the ability of a speaker to mould discussion of an issue towards their own framing

- How do speakers attribute importance to one another?
  1. Explicitly
  2. Implicitly
- Adopting the framing of an issue implies a perspective is important
- Influential MP's 'shape the debate' by changing the vocabulary of other MPs
- Implication: infer influence from patterns of word use in debate

Penalised regression for modelling influence

- Represent each speech as a vector of word proportions, w_i
- Assumption: If w_i is predictive of w_j, then i influences j
- Problem: relationship between i and j might be confounded by h
- Solution: model each speech as a function of all previous speeches

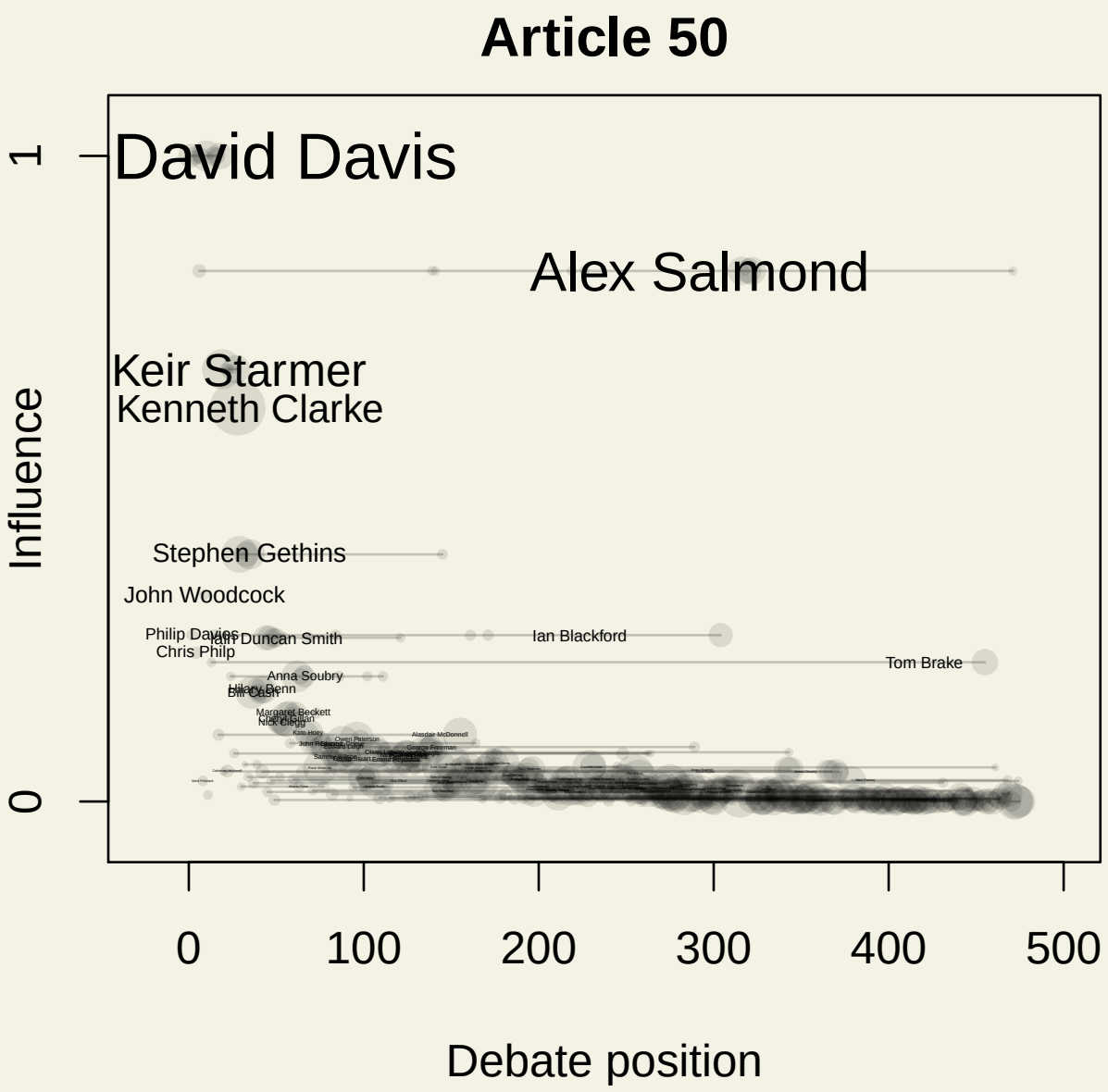
$$w_j = \sum_{s=1}^{j-1} \beta_s w_s + \epsilon_j$$

- β_i captures the influence of i on j conditional on all other speeches
- Penalise β_s via LASSO $\rightarrow$ 'hard' for one speech to influence another
- Rank MPs by the *quantity* and *quality* of those they influence

Data

- All debates in the UK House of Commons between 1979 and 2018
- $\approx$ 60,000 debates
- $\approx$ 1.5 million speeches
- For each debate, score the relative influence of each MP via LASSO

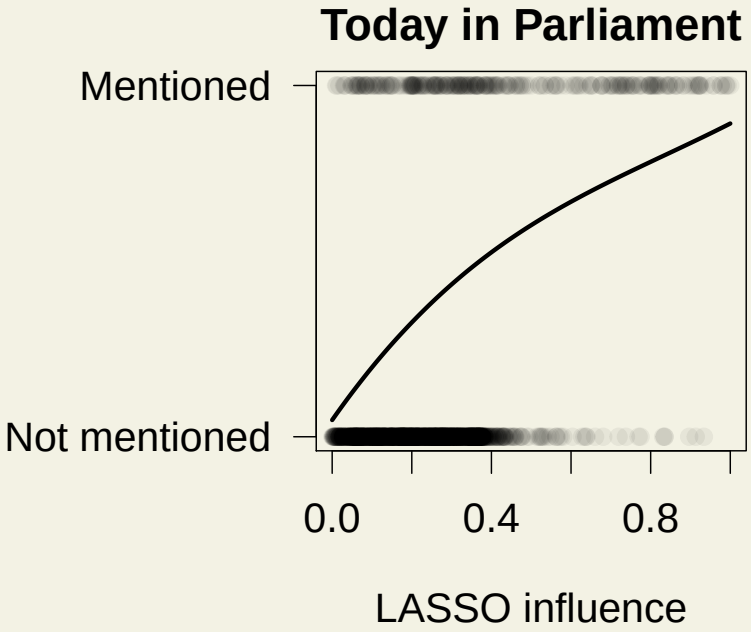
Example debate



Debate-level validation

Do influential **speeches** receive more media coverage?

- BBC's 'Today in Parliament' radio programme
- Excerpts from the most important speeches each day
- 17 days in 2018, 78 debates, 1749 speaker-debate observations



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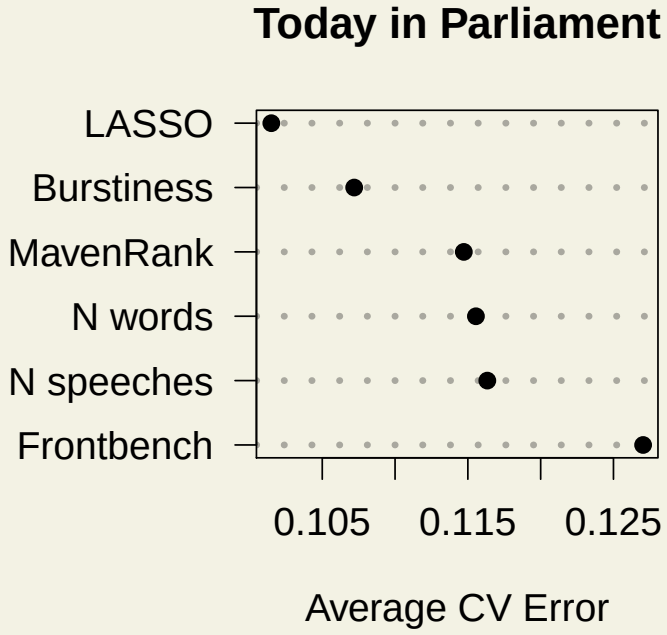
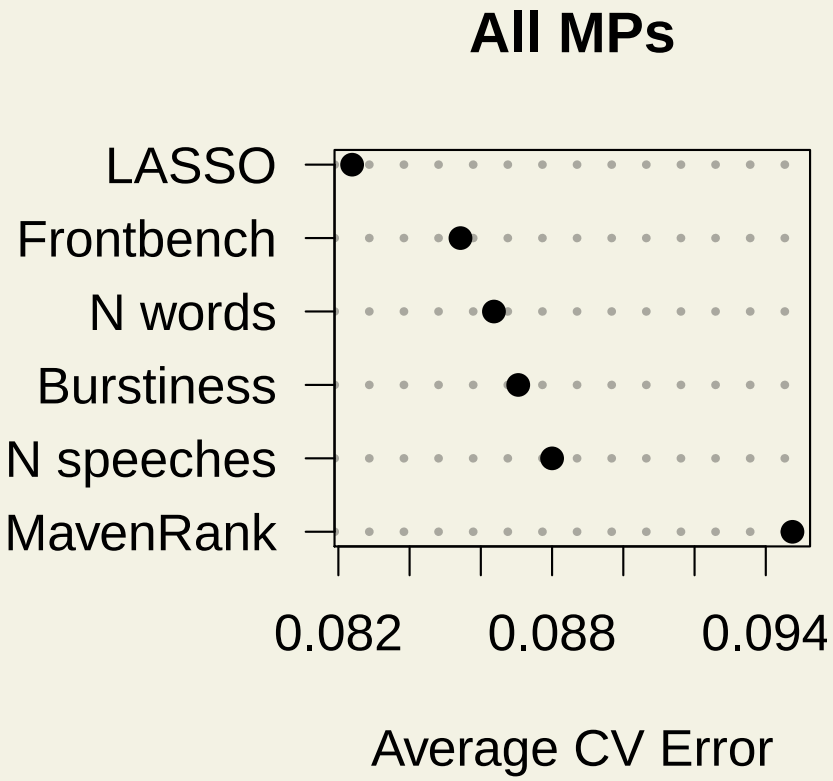


Individual-level validation

Do influential **speakers** receive more media coverage?

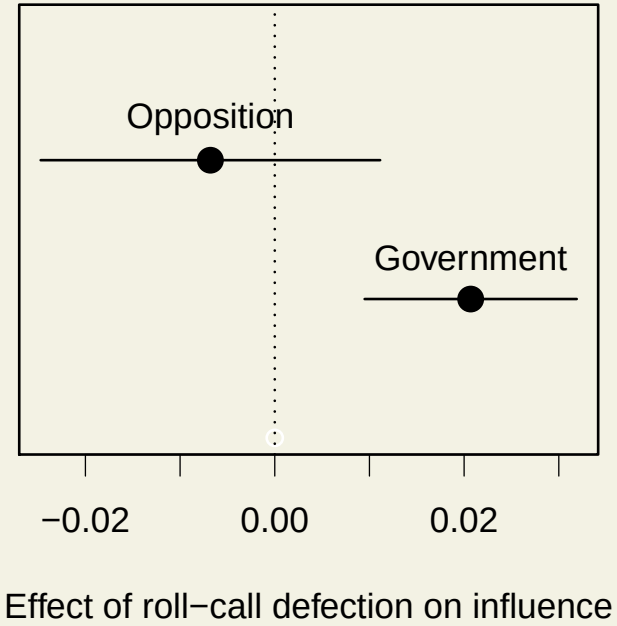
- For each year, measure 'power' of MPs from newspaper mentions (Ban et al, 2017)
- 19,000 Guardian articles from 2010 to 2017
- For each year, measure relative influence in debate of MPs
- Correlation of 0.42 between influence and Guardian mentions



Influence and roll-call defections

Are government rebels influential?

- Government rebels are more likely to give speeches to signal opposition (Slapin et al, 2018)
- Match debates to parliamentary votes: 42,000 speaker-debate observations
- Government rebels are $\approx 10\%$ more influential than loyalists



Future work

- Methodological
  - Uncertainty
  - Parameter tuning in LASSO regression
- Substantive
  - Better applications
  - Other applications